



A Sound and Complete Equational Theory for 3-Qubit Toffoli-Hadamard Circuits

Matthew Amy



Neil J. Ross



Scott Wesley



Introduction

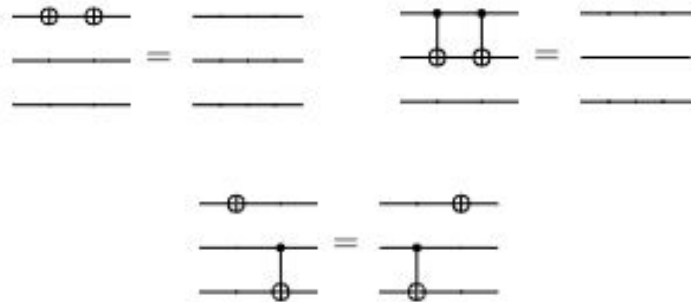
Toffoli-Hadamard and Equational Theories



A Quick Review: What Is An Equational Theory?

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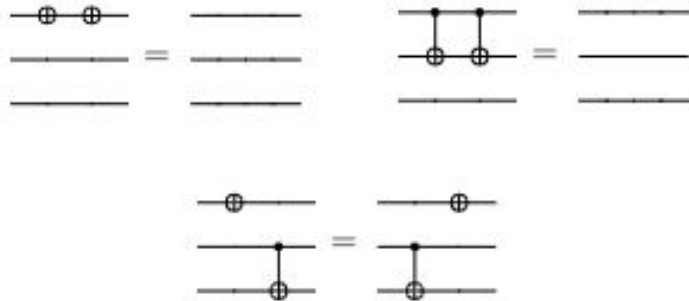
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For a fixed number of qubits, the circuits form a monoid, which means that such relations are explicit.

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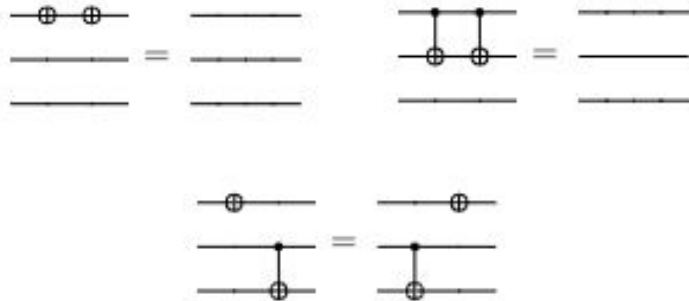
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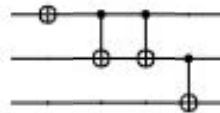
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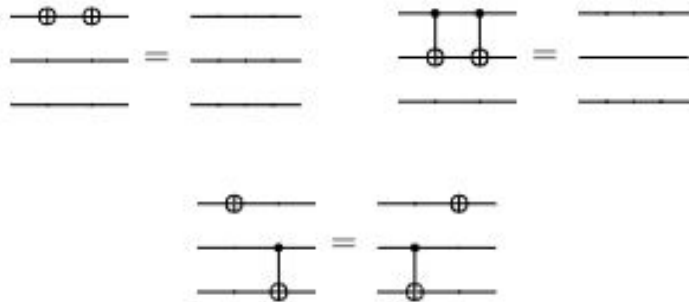


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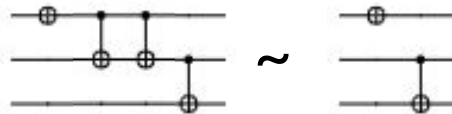


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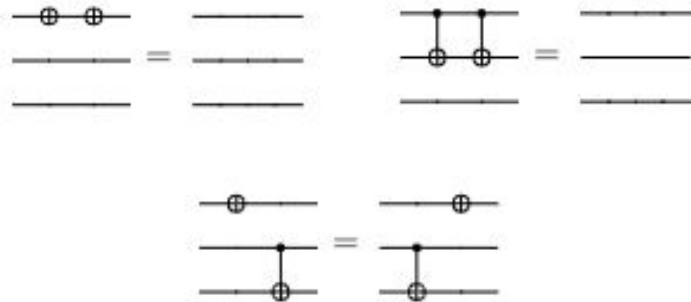


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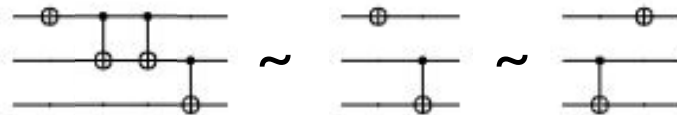


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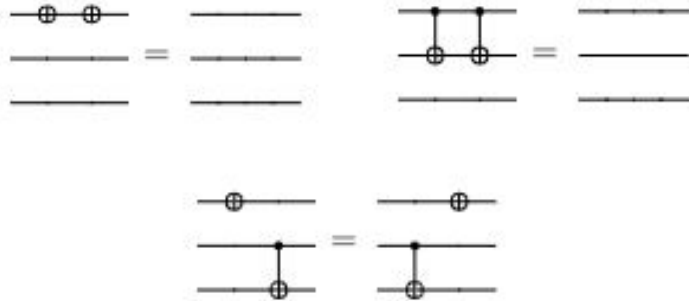


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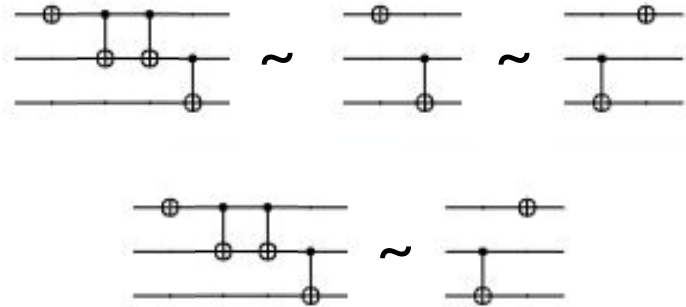


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In an equational theory, we also have a semantic interpretation for the generators, say $[[--]]$.



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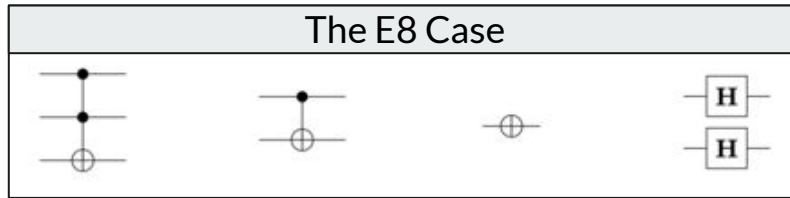
Completeness

$$[[u]] = [[v]] \implies u \sim v$$

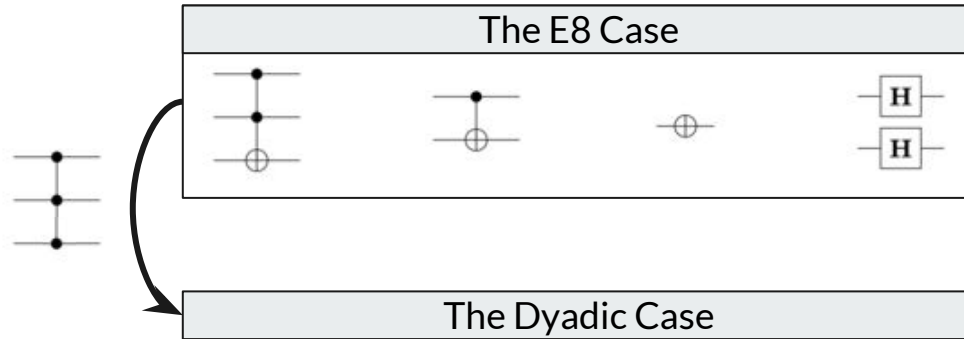


Towards a Toffoli-Hadamard Equational Theory

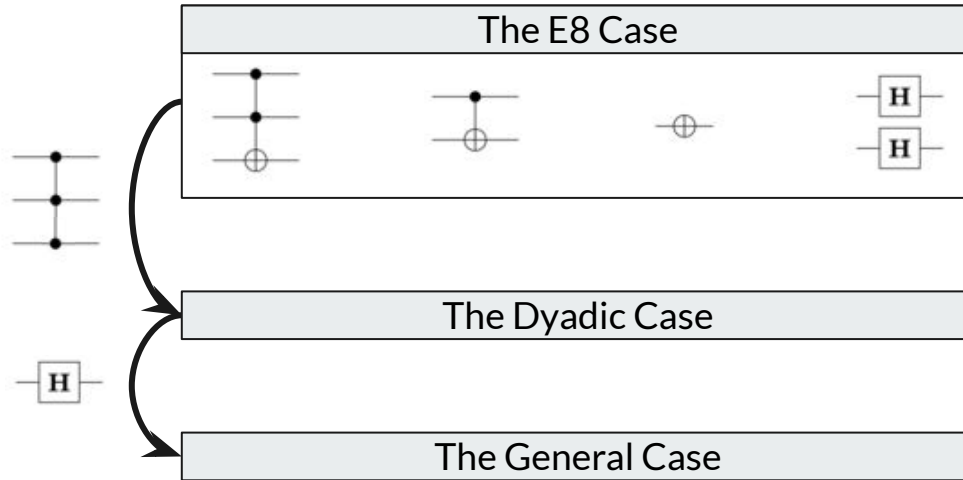
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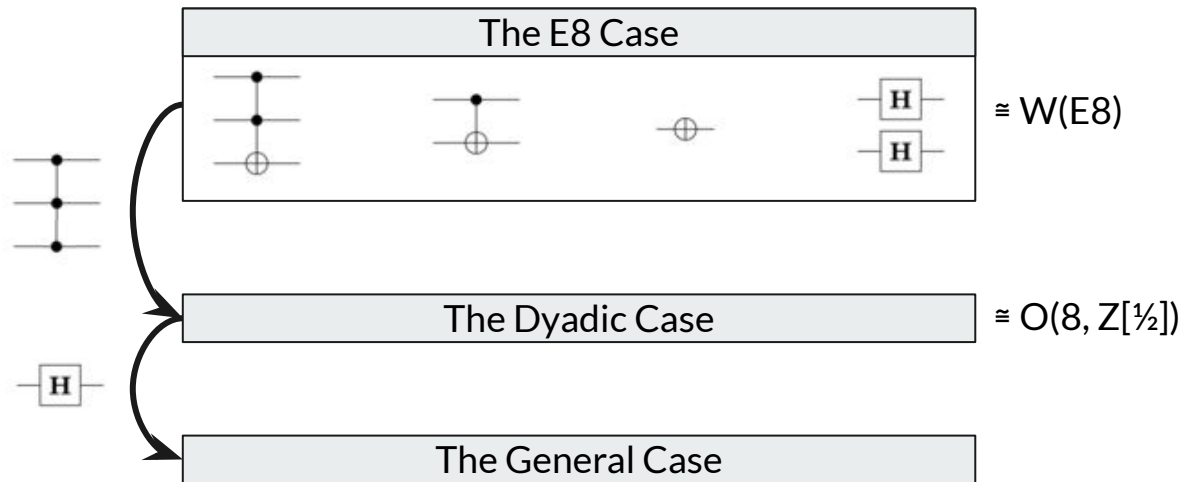
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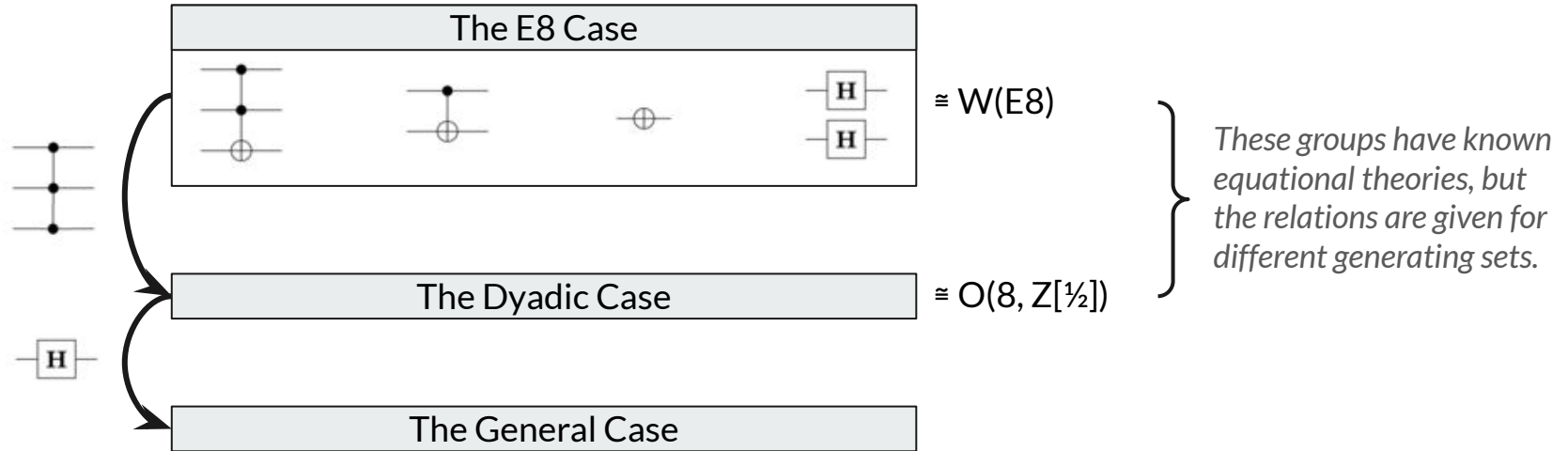
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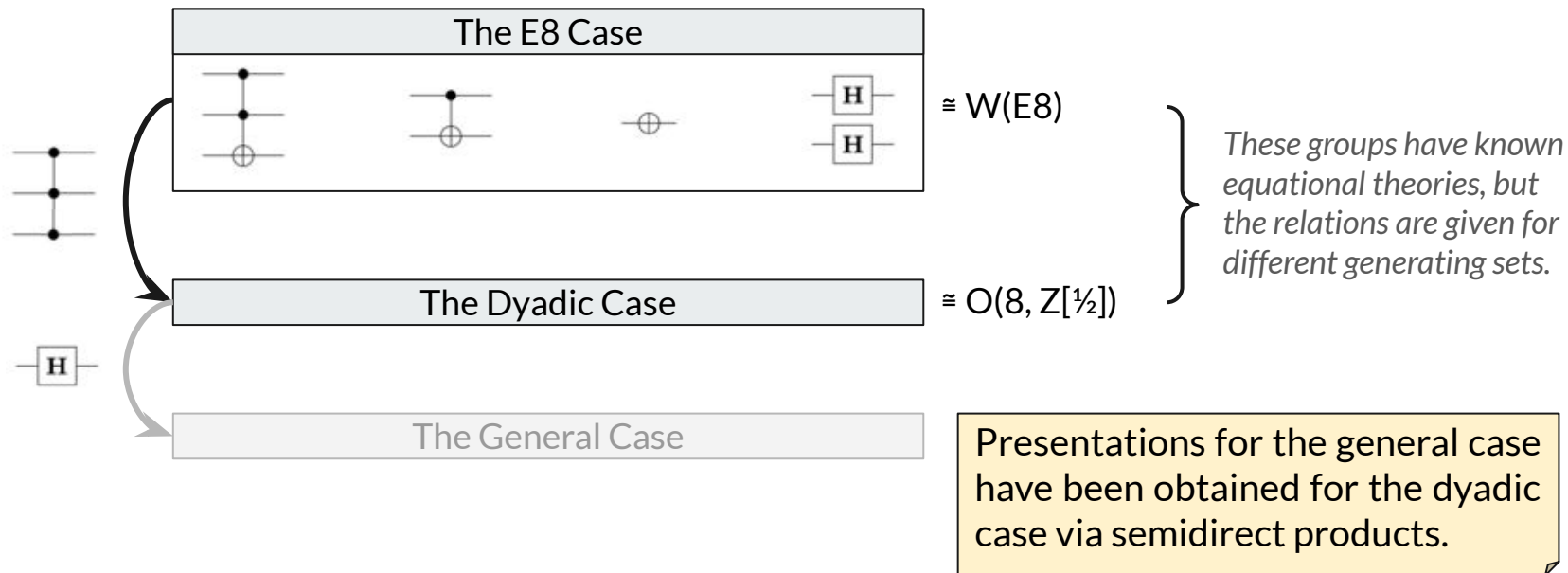
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Key Technique: Tietze Transformations



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If x is not in Σ , and w is a word over Σ , then x can be added to Σ with the new relation $x \sim w$.
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Tietze Transformations and Semantics

Context: Starting from a Complete Theory

Tietze transformations establish isomorphisms between presentations. It is often the case that R is already complete with respect to some $[[-]]$.



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Since R is complete for $[[-]]$, then to prove that $u \sim v$, it suffices to show that $[[u]] = [[v]]$. That is, $\text{Rel}(+)$ does not require derivational proofs.



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Semantic Gen(+): Preserving Completeness

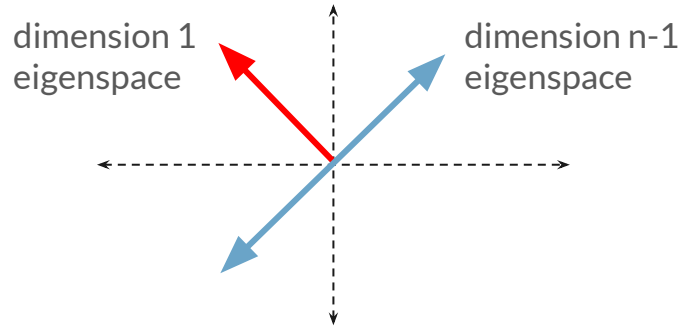
Assume that g is a gate with interpretation $[[g]]$. Then the Gen(+) transformation for $g \sim w$ will preserve completeness if and only if $[[g]] = [[w]]$.

The E8 Case

From Coxeter Geometry to Circuit Rewriting

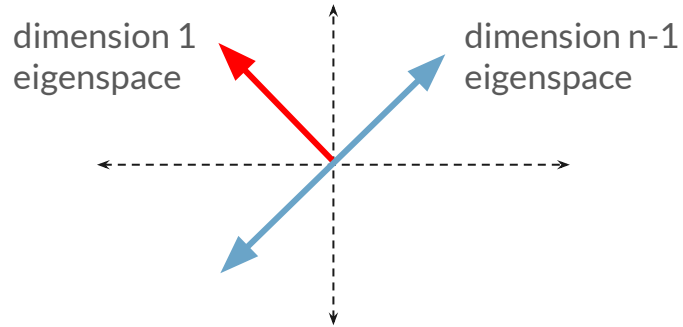
Reflection Groups: An Example in 2-Dimensions

What is a Reflection?

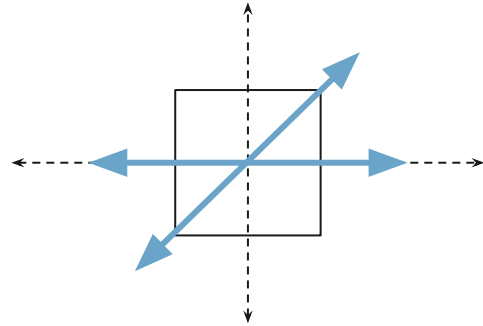


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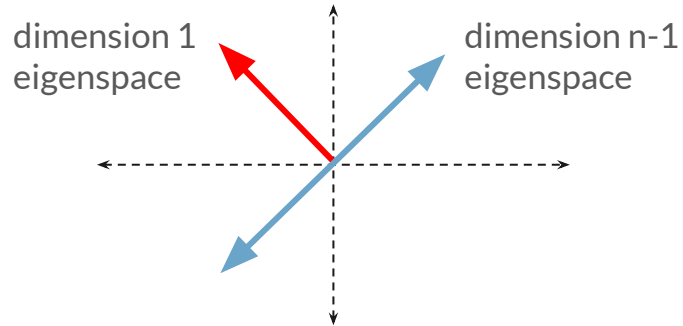


Dihedral Groups as Reflection Groups



Coxeter Presentations for Finite Reflection Groups

What is a Reflection?



Equational Theories for Finite Refl Groups

For each refl. r , introduce a relation:

$$r^2 = 1$$

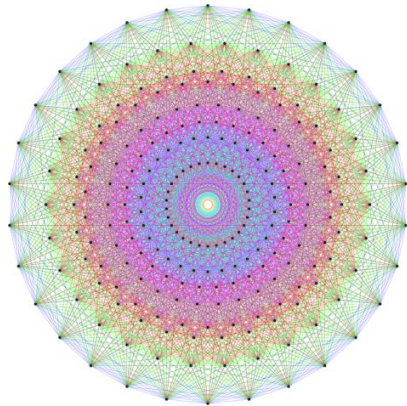
For each pair of refls. r, s , introduce a relation:

$$(rs)^n = 1$$

Given n reflections, there will be n^2 relations.

W(E8): The Automorphisms of the E8-Lattice

The E8-Lattice

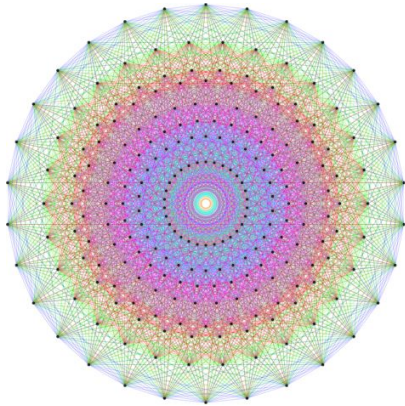


The Automorphisms as a Coxeter Group

$$\begin{bmatrix} 1 & 3 & 2 & 2 & 2 & 2 & 2 & 2 \\ 3 & 1 & 3 & 2 & 2 & 2 & 2 & 2 \\ 2 & 3 & 1 & 3 & 2 & 2 & 2 & 2 \\ 2 & 2 & 3 & 1 & 3 & 2 & 2 & 2 \\ 2 & 2 & 2 & 3 & 1 & 3 & 3 & 2 \\ 2 & 2 & 2 & 2 & 3 & 1 & 2 & 2 \\ 2 & 2 & 2 & 2 & 3 & 2 & 1 & 3 \\ 2 & 2 & 2 & 2 & 2 & 2 & 3 & 1 \end{bmatrix}$$

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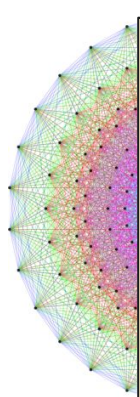
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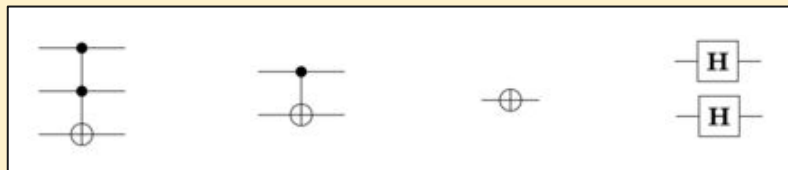
$(r_1 r_7)^2 = 1$

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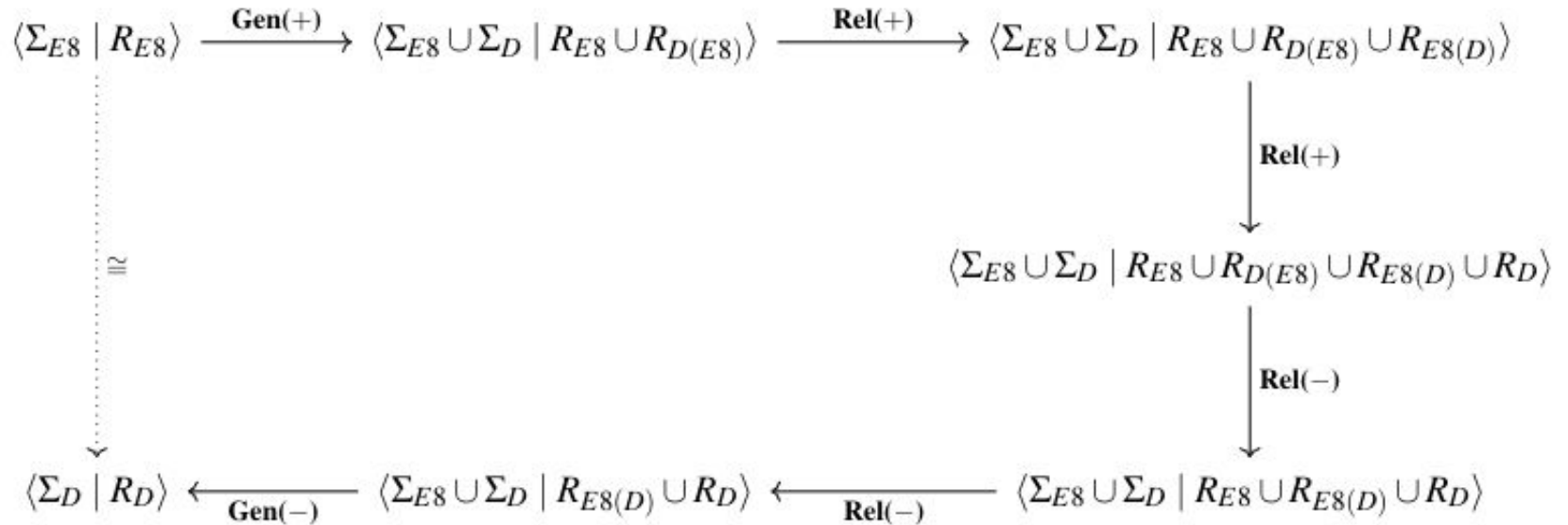
The standard representation of this group includes all E8 circuit generators



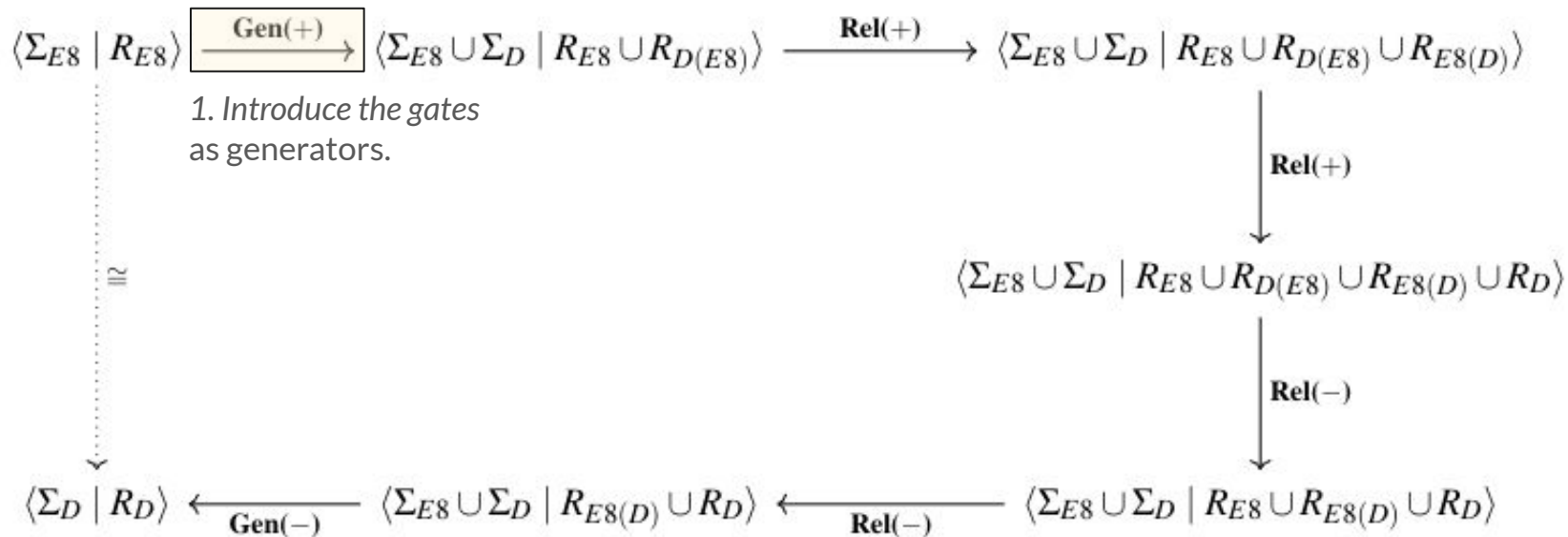
The Automorphisms as a Coxeter Group

$$\begin{bmatrix} 2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \\ 3 & 3 & 2 \\ 1 & 2 & 2 \\ 2 & 1 & 3 \\ 2 & 3 & 1 \end{bmatrix}$$

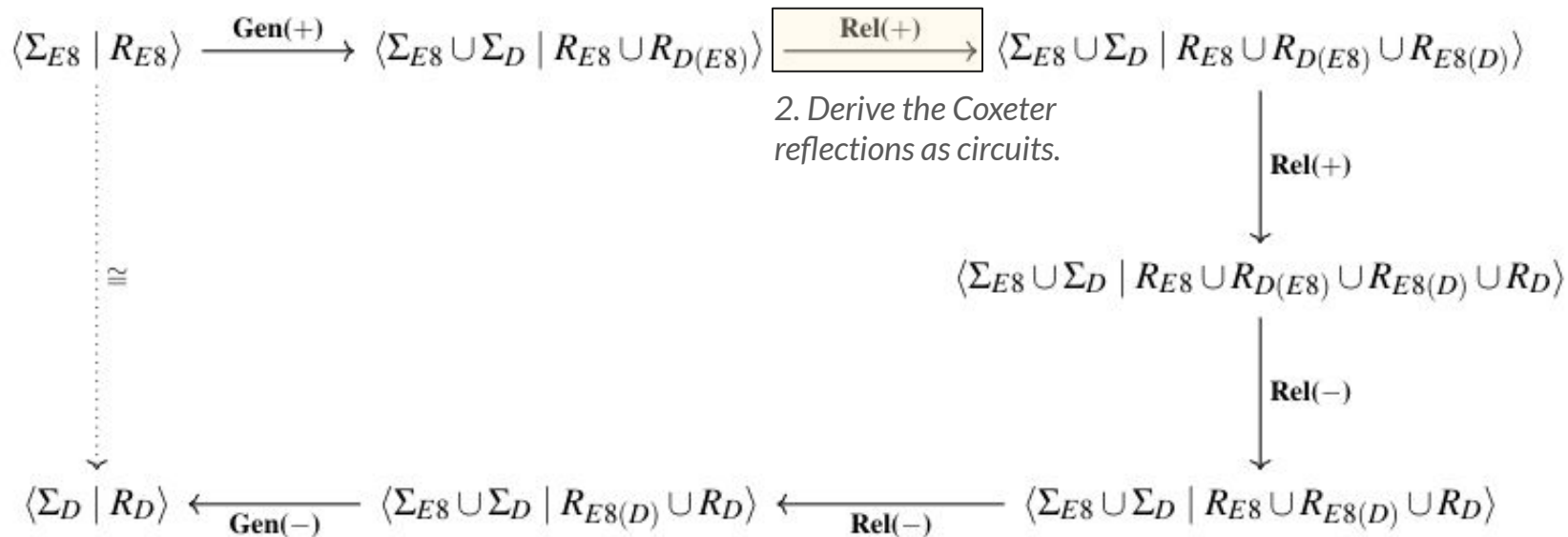
From W(E8) to the E8 Gate Set



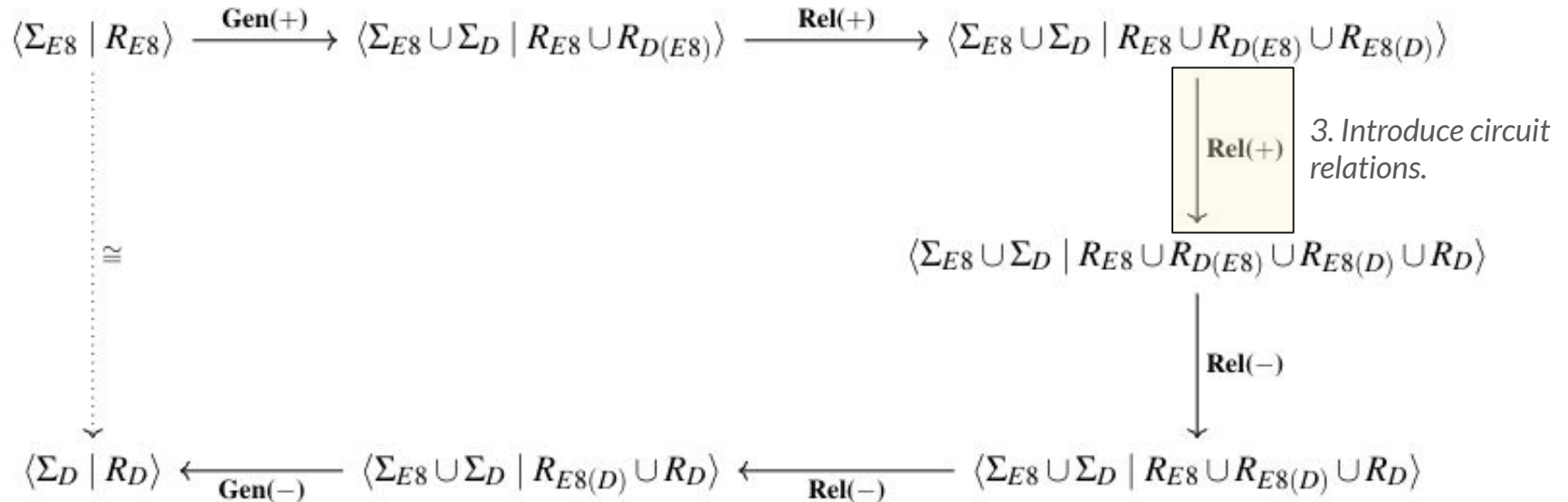
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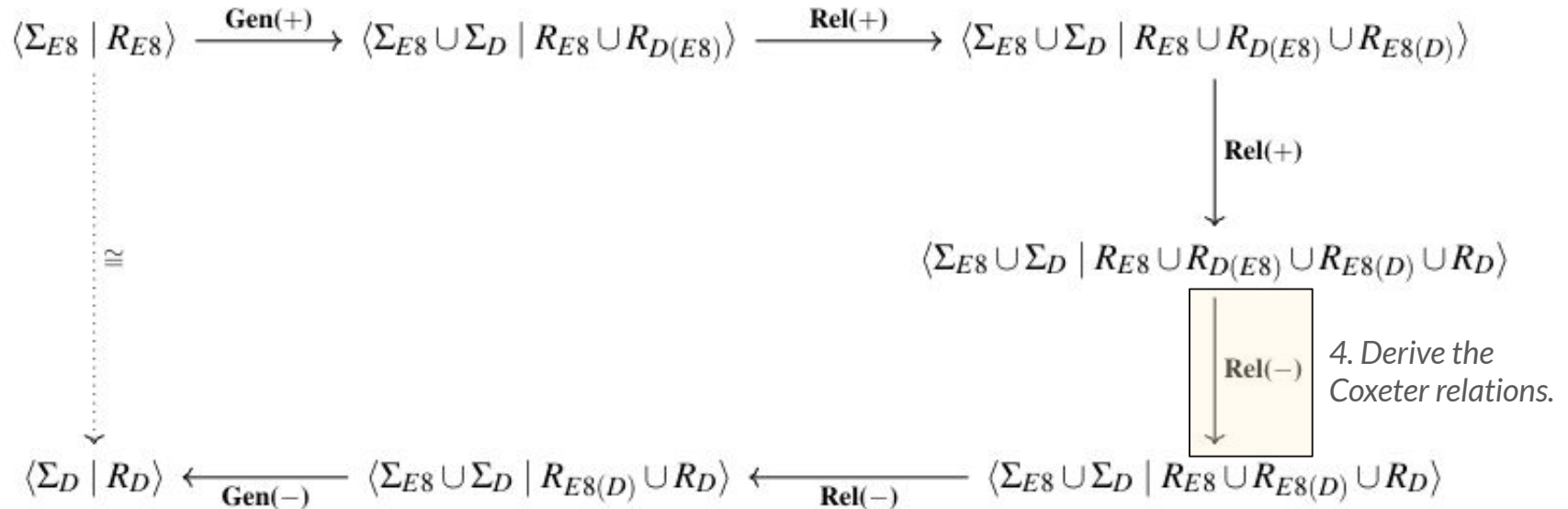
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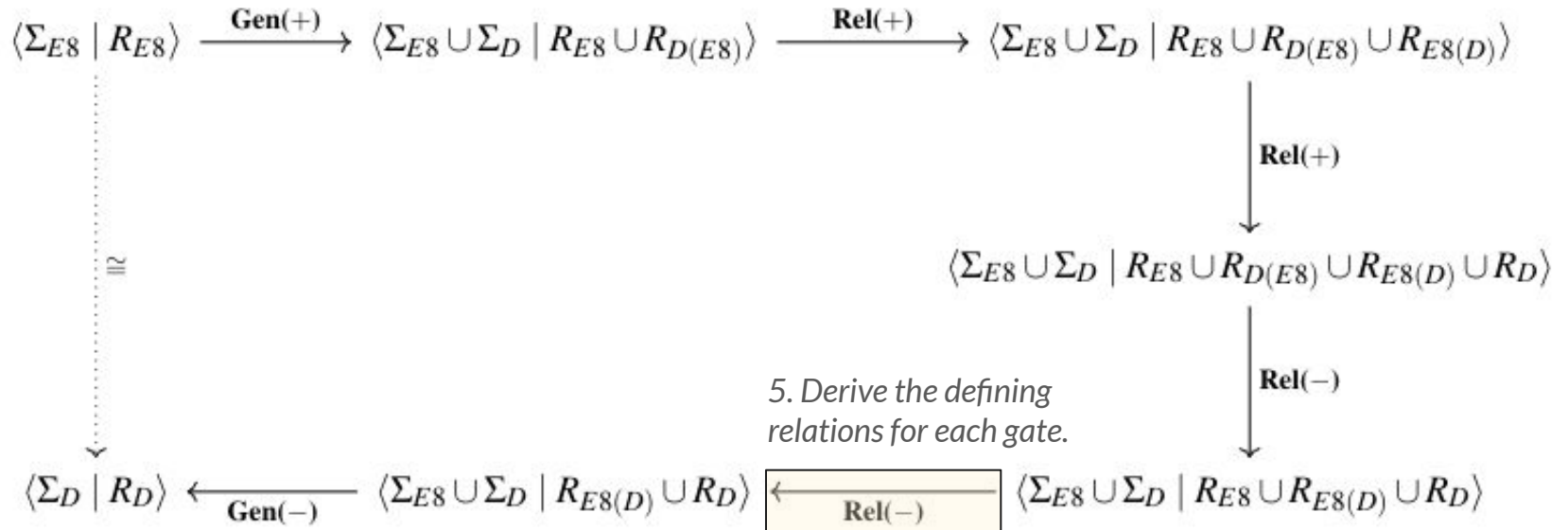
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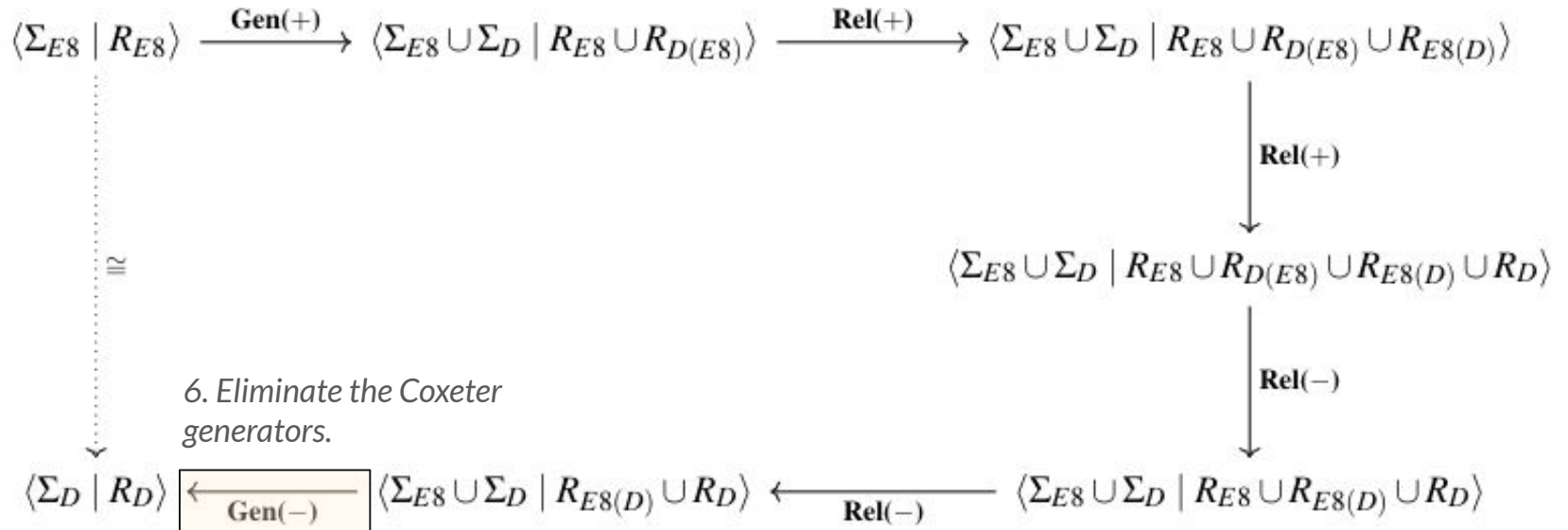
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$$\langle \Sigma_{E8} \mid R_{E8} \rangle \xrightarrow{\text{Gen}(+)} \langle \Sigma_{E8} \cup \Sigma_D \mid R_{E8} \cup R_{D(E8)} \rangle \xrightarrow{\text{Rel}(+)} \langle \Sigma_{E8} \cup \Sigma_D \mid R_{E8} \cup R_{D(E8)} \cup R_{E8(D)} \rangle$$

All transformations have been machine-verified in a tool we are developing called **Tietze**. Tietze provides a language to express rewriting proofs in finitely presentable groups, with tools to verify the soundness of each relation and the correctness of each derivation.

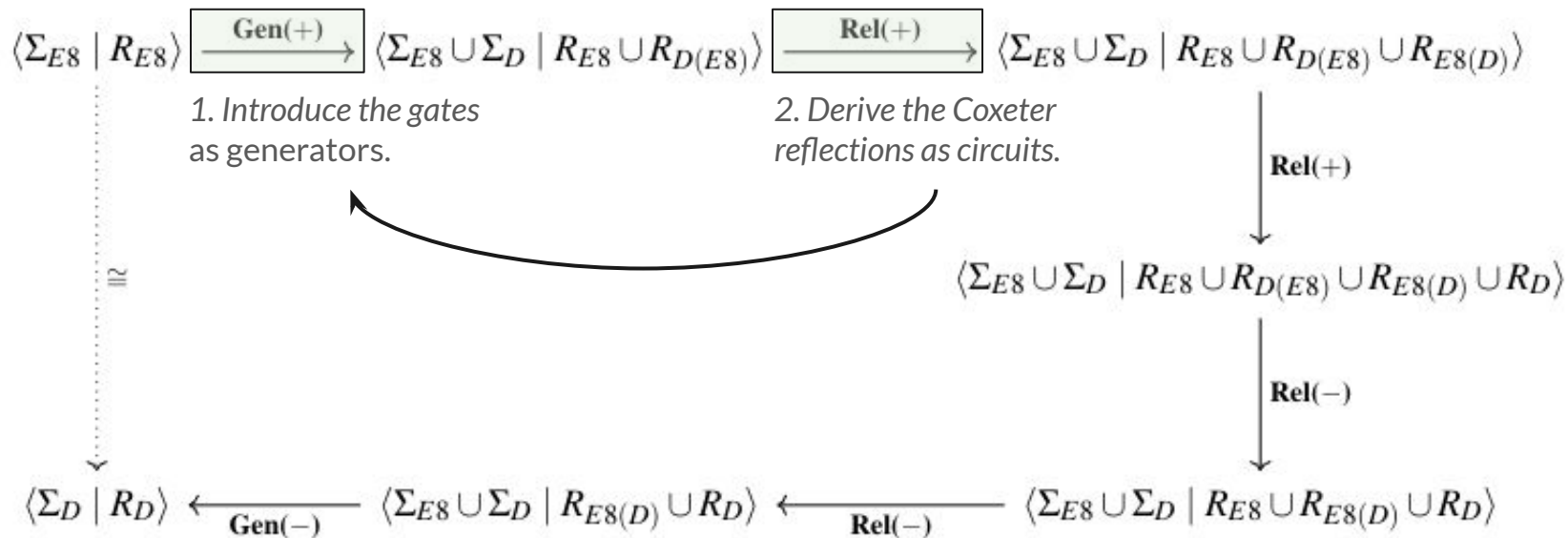
$$\downarrow \text{Rel}(+)$$

$$\langle \Sigma_{E8} \cup \Sigma_D \mid R_{E8} \cup R_{D(E8)} \cup R_{E8(D)} \cup R_D \rangle$$

$$\downarrow \text{Rel}(-)$$

$$\langle \Sigma_D \mid R_D \rangle \xleftarrow{\text{Gen}(-)} \langle \Sigma_{E8} \cup \Sigma_D \mid R_{E8(D)} \cup R_D \rangle \xleftarrow{\text{Rel}(-)} \langle \Sigma_{E8} \cup \Sigma_D \mid R_{E8} \cup R_{E8(D)} \cup R_D \rangle$$

From W(E8) to the E8 Gate Set





Deriving Circuits for the Coxeter Reflections



Deriving Circuits for the Coxeter Reflections

1. Start From a Known Refl (e.g., CCX)

Invariant Subspace: The CCX gate fixes $|00-\rangle$, $|00+\rangle$, $|01-\rangle$, $|01+\rangle$, $|10-\rangle$, $|10+\rangle$, and $|11+\rangle$.

(-1)-Eigenspace: CCX sends $|11-\rangle$ to $-|11-\rangle$, because X send $|-\rangle$ to $-|-\rangle$.

This means that CCX is a reflection about the normal vector $|11-\rangle$.

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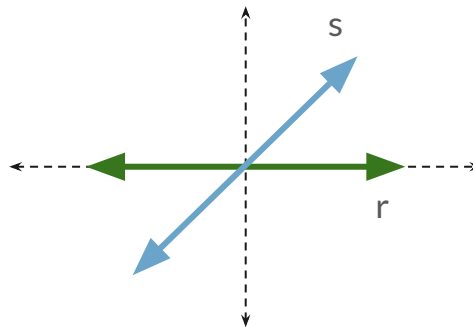
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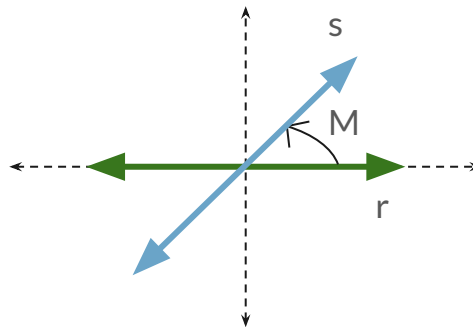
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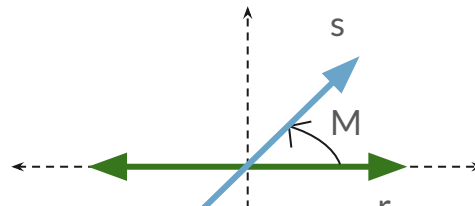
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It follows geometrically that:
$$r = M^{-1}sM$$

The Dyadic Case

A Meet-in-the-Middle Approach

From W(E8) to O(8, Z[1/2])

$$\begin{aligned}
 CZ_{0,1} &\approx K_{1,2} \cdot CX_{0,1} \cdot K_{1,2} & (1) \\
 X_1 &\approx CX_{0,1} \cdot X_0 \cdot CX_{0,1} \cdot X_0 & (2) \\
 Z_0 &\approx CZ_{0,1} \cdot CX_{0,1} \cdot CZ_{0,1} \cdot CX_{0,1} & (3) \\
 Z_1 &\approx K_{1,2} \cdot X_1 \cdot K_{1,2} & (4) \\
 CX_{2,0} &\approx X_1 \cdot CCX_{1,2} \cdot X_1 \cdot CCX_{1,2} & (5) \\
 CX_{2,1} &\approx CX_{2,0} \cdot CX_{0,1} \cdot CX_{2,0} \cdot CX_{0,1} & (6) \\
 CX_{1,2} &\approx K_{1,2} \cdot CX_{2,1} \cdot K_{1,2} & (7) \\
 \sigma_{1,2} &\approx CX_{1,2} \cdot CX_{2,1} \cdot CX_{1,2} & (8) \\
 CX_{0,1} &\approx \sigma_{1,2} \cdot CX_{0,1} \cdot \sigma_{1,2} & (9) \\
 \sigma_{0,2} &\approx CX_{0,2} \cdot CX_{2,0} \cdot CX_{0,2} & (10) \\
 K_{0,1} &\approx \sigma_{0,2} \cdot K_{1,2} \cdot \sigma_{0,2} & (11) \\
 CX_{1,0} &\approx K_{0,1} \cdot CX_{0,1} \cdot K_{0,1} & (12) \\
 \sigma_{0,1} &\approx CX_{0,1} \cdot CX_{1,0} \cdot CX_{0,1} & (13) \\
 CCX_{0,1} &\approx \sigma_{0,1} \cdot CCX_{1,2} \cdot \sigma_{0,1} & (14) \\
 X_2 &\approx \sigma_{0,2} \cdot X_0 \cdot \sigma_{0,2} & (15) \\
 Z_2 &\approx \sigma_{0,2} \cdot Z_0 \cdot \sigma_{0,2} & (16) \\
 CCX_{0,1} &\approx K_{1,2} \cdot CCX_{0,2} \cdot K_{1,2} & (17) \\
 CZ_{0,2} &\approx \sigma_{1,2} \cdot CZ_{0,1} \cdot \sigma_{1,2} & (18) \\
 CZ_{1,2} &\approx \sigma_{0,1} \cdot CZ_{0,2} \cdot \sigma_{0,1} & (19) \\
 \sigma_{1,2} &\approx CZ_{1,2} \cdot K_{1,2} \cdot CZ_{1,2} \cdot K_{1,2} \cdot CZ_{1,2} \cdot K_{1,2} & (20) \\
 X_0 \cdot X_0 &\approx \epsilon & (21) \\
 CX_{0,1} \cdot CX_{0,1} &\approx \epsilon & (22) \\
 K_{1,2} \cdot K_{1,2} &\approx \epsilon & (23) \\
 CCX_{1,2} \cdot CCX_{1,2} &\approx \epsilon & (24) \\
 K_{0,1} \cdot K_{0,1} &\approx \epsilon & (25) \\
 CX_{1,2} \cdot X_0 &\approx X_0 \cdot CX_{1,2} & (26) \\
 X_0 \cdot K_{1,2} &\approx K_{1,2} \cdot X_0 & (27) \\
 X_1 &\approx \sigma_{0,1} \cdot X_0 \cdot \sigma_{0,1} & (28) \\
 CX_{2,0} &\approx \sigma_{0,2} \cdot CX_{0,2} \cdot \sigma_{0,2} & (29) \\
 CX_{1,2} &\approx \sigma_{0,1} \cdot CX_{0,2} \cdot \sigma_{0,1} & (30) \\
 CX_{2,1} &\approx \sigma_{0,1} \cdot CX_{2,0} \cdot \sigma_{0,1} & (31) \\
 CCX_{0,1} &\approx \sigma_{0,1} \cdot CCX_{1,2} \cdot \sigma_{0,1} & (32) \\
 CCX_{0,1} &\approx \sigma_{1,2} \cdot CCX_{0,2} \cdot \sigma_{1,2} & (33) \\
 Z_1 &\approx \sigma_{0,1} \cdot Z_0 \cdot \sigma_{0,1} & (34) \\
 K_{0,1} &\approx \sigma_{0,1} \cdot K_{0,1} \cdot \sigma_{0,1} & (35) \\
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 X_0 \cdot CCX_{1,2} &\approx CCX_{1,2} \cdot X_0 & (37) \\
 X_0 \cdot CX_{1,0} &\approx CX_{1,0} \cdot X_0 & (38) \\
 K_{0,1} \cdot K_{1,2} &\approx K_{1,2} \cdot K_{0,1} & (39) \\
 CZ_{0,1} \cdot CZ_{1,2} &\approx CZ_{1,2} \cdot CZ_{0,1} & (40) \\
 K_{0,1} \cdot Z_0 &\approx X_0 \cdot K_{0,1} & (41) \\
 X_0 \cdot CCX_{0,1} &\approx CCX_{0,1} \cdot CX_{1,2} \cdot X_0 & (42) \\
 CX_{0,1} \cdot CZ_{1,2} &\approx CZ_{1,2} \cdot CZ_{0,2} \cdot CX_{0,1} & (43) \\
 CX_{1,2} \cdot CCX_{1,2} &\approx CCX_{1,2} \cdot CX_{1,0} \cdot CX_{1,2} & (44) \\
 CCX_{1,2} \cdot CX_{0,1} &\approx CX_{0,1} \cdot CCX_{0,2} \cdot CCX_{1,2} \cdot CCX_{0,2} & (45) \\
 CCX_{0,1} \cdot CCX_{0,2} &\approx CCX_{0,2} \cdot CCX_{0,1} \cdot CCX_{0,2} \cdot CCX_{0,1} & (46)
 \end{aligned}$$

$$X_{[\mu,b]}^2 \approx \epsilon \quad (1a)$$

$$(-1)_{[\mu]}^2 \approx \epsilon \quad (1b)$$

$$K_{[\mu,b,c,d]}^2 \approx \epsilon \quad (1c)$$

$$X_{[\mu,b]} X_{[\nu,c]} \approx X_{[\nu,c]} X_{[\mu,b]} \quad (2a)$$

$$X_{[\mu,b]} (-1)_{[\nu]} \approx (-1)_{[\nu]} X_{[\mu,b]} \quad (2b)$$

$$X_{[\mu,b]} K_{[\nu,c,d,e]} \approx K_{[\nu,c,d,e]} X_{[\mu,b]} \quad (2c)$$

$$(-1)_{[\mu]} (-1)_{[\nu]} \approx (-1)_{[\nu]} (-1)_{[\mu]} \quad (2d)$$

$$(-1)_{[\mu]} K_{[\nu,c,d,e]} \approx K_{[\nu,c,d,e]} (-1)_{[\mu]} \quad (2e)$$

$$K_{[\mu,b,c,d]} K_{[\nu,f,g,h]} \approx K_{[\nu,f,g,h]} K_{[\mu,b,c,d]} \quad (2f)$$

$$X_{[\mu,c]} X_{[\nu,b]} \approx X_{[\nu,b]} X_{[\mu,c]} \quad (3a)$$

$$X_{[\mu,b]} X_{[\nu,c]} \approx X_{[\nu,c]} X_{[\mu,b]} \quad (3b)$$

$$X_{[\mu,b]} (-1)_{[\nu]} \approx (-1)_{[\nu]} X_{[\mu,b]} \quad (3c)$$

$$X_{[\mu,c]} K_{[\nu,b,d,e]} \approx K_{[\nu,b,d,e]} X_{[\mu,c]} \quad (3d)$$

$$X_{[\nu,b]} K_{[\mu,a,b,c,d]} \approx K_{[\mu,a,b,c,d]} X_{[\nu,b]} \quad (3e)$$

$$X_{[\nu,c]} K_{[\mu,b,c,d]} \approx K_{[\mu,b,c,d]} X_{[\nu,c]} \quad (3f)$$

$$X_{[\mu,d]} K_{[\nu,a,b,c]} \approx K_{[\nu,a,b,c]} X_{[\mu,d]} \quad (3g)$$

$$X_{[\mu,b]} K_{[\mu,b,c,d]} \approx K_{[\mu,b,c,d]} X_{[\mu,b]} (-1)_{[\mu]} (-1)_{[\mu]} \quad (4a)$$

$$X_{[\nu,c]} K_{[\mu,a,b,c,d]} \approx (-1)_{[\mu]} K_{[\mu,a,b,c,d]} (-1)_{[\mu]} K_{[\mu,a,b,c,d]} (-1)_{[\nu]} \quad (4b)$$

$$X_{[\nu,c]} K_{[\nu,a,b,c,d]} \approx K_{[\nu,a,b,c,d]} X_{[\nu,c]} \quad (4c)$$

$$K_{[\mu,b,c,d]} K_{[\nu,d,e,f]} \approx K_{[\nu,d,e,f]} K_{[\mu,b,c,d]} \quad (5a)$$

$$\begin{aligned}
 (-1)_{[\mu]} (-1)_{[\nu]} X_{[\mu,c]} K_{[\nu,f,g,h]} K_{[\mu,b,c,d]} X_{[\mu,d,e]} K_{[\mu,b,c,d]} K_{[\nu,f,g,h]} X_{[\mu,c]} (-1)_{[\mu]} (-1)_{[\nu]} \\
 \approx
 \end{aligned} \quad (6a)$$

$$K_{[\nu,f,g,h]} K_{[\mu,b,c,d]} X_{[\mu,d,e]} K_{[\mu,b,c,d]} K_{[\nu,f,g,h]} \quad (6b)$$

From W(E8) to O(8, Z[1/2])

From the following section, we have 48 circuit relations for W(E8).

$$\begin{aligned}
 CX_{1,2} &\approx \sigma_{1,2} \cdot \sigma_{2,1} \cdot \sigma_{1,2} & (7) \\
 \sigma_{1,2} &\approx CX_{1,2} \cdot CX_{2,1} \cdot CX_{1,2} & (8) \\
 CX_{0,1} &\approx \sigma_{1,2} \cdot CX_{0,1} \cdot \sigma_{1,2} & (9) \\
 \sigma_{0,1} &\approx CX_{0,2} \cdot CX_{2,0} \cdot CX_{0,2} & (10) \\
 K_{0,1} &\approx \sigma_{0,2} \cdot K_{1,2} \cdot \sigma_{0,2} & (11) \\
 CX_{1,0} &\approx K_{0,1} \cdot CX_{0,1} \cdot K_{0,1} & (12) \\
 \sigma_{0,1} &\approx CX_{0,1} \cdot CX_{1,0} \cdot CX_{0,1} & (13) \\
 CCX_{0,1} &\approx \sigma_{0,1} \cdot CCX_{1,2} \cdot \sigma_{0,1} & (14) \\
 X_2 &\approx \sigma_{0,2} \cdot X_0 \cdot \sigma_{0,2} & (15) \\
 Z_2 &\approx \sigma_{0,2} \cdot Z_0 \cdot \sigma_{0,2} & (16) \\
 CCX_{0,1} &\approx K_{1,2} \cdot CCX_{0,2} \cdot K_{1,2} & (17) \\
 CZ_{0,1} &\approx \sigma_{1,2} \cdot CZ_{0,1} \cdot \sigma_{1,2} & (18) \\
 CZ_{1,2} &\approx \sigma_{0,1} \cdot CZ_{0,2} \cdot \sigma_{0,1} & (19) \\
 \sigma_{1,2} &\approx CZ_{1,2} \cdot K_{1,2} \cdot CZ_{1,2} \cdot K_{1,2} \cdot CZ_{1,2} \cdot K_{1,2} & (20) \\
 X_0 \cdot X_0 &\approx \epsilon & (21) \\
 CX_{0,1} \cdot CX_{0,1} &\approx \epsilon & (22) \\
 K_{1,2} \cdot K_{1,2} &\approx \epsilon & (23) \\
 CX_{1,2} &\approx \epsilon & (24) \\
 K_{0,1} &\approx \epsilon & (25) \\
 X_0 &\approx X_0 \cdot CX_{1,2} & (26) \\
 K_{1,2} &\approx K_{1,2} \cdot X_0 & (27) \\
 X_1 &\approx \sigma_{0,1} \cdot X_0 \cdot \sigma_{0,1} & (28) \\
 CX_{2,0} &\approx \sigma_{0,2} \cdot CX_{0,2} \cdot \sigma_{0,2} & (29) \\
 CX_{1,2} &\approx \sigma_{0,1} \cdot CX_{0,2} \cdot \sigma_{0,1} & (30) \\
 CX_{2,1} &\approx \sigma_{0,1} \cdot CX_{2,0} \cdot \sigma_{0,1} & (31) \\
 CCX_{0,1} &\approx \sigma_{0,1} \cdot CCX_{1,2} \cdot \sigma_{0,1} & (32) \\
 CCX_{0,1} &\approx \sigma_{1,2} \cdot CCX_{0,2} \cdot \sigma_{1,2} & (33) \\
 Z_1 &\approx \sigma_{0,1} \cdot Z_0 \cdot \sigma_{0,1} & (34) \\
 K_{0,1} &\approx \sigma_{0,1} \cdot K_{0,1} \cdot \sigma_{0,1} & (35) \\
 CCX_{1,2} \cdot CX_{1,0} &\approx CX_{1,0} \cdot CCX_{1,2} & (36) \\
 X_0 \cdot CCX_{1,2} &\approx CCX_{1,2} \cdot X_0 & (37) \\
 X_0 \cdot CX_{1,0} &\approx CX_{1,0} \cdot X_0 & (38) \\
 K_{0,1} \cdot K_{1,2} &\approx K_{1,2} \cdot K_{0,1} & (39) \\
 CZ_{0,1} \cdot CZ_{1,2} &\approx CZ_{1,2} \cdot CZ_{0,1} & (40) \\
 K_{0,1} \cdot Z_0 &\approx X_0 \cdot K_{0,1} & (41) \\
 X_0 \cdot CCX_{0,1} &\approx CCX_{0,1} \cdot CX_{1,2} \cdot X_0 & (42) \\
 CX_{0,1} \cdot CZ_{1,2} &\approx CZ_{1,2} \cdot CZ_{0,2} \cdot CX_{0,1} & (43) \\
 CX_{1,2} \cdot CCX_{1,2} &\approx CCX_{1,2} \cdot CX_{1,0} \cdot CX_{1,2} & (44) \\
 CCX_{1,2} \cdot CX_{0,1} &\approx CX_{0,1} \cdot CCX_{0,2} \cdot CCX_{1,2} \cdot CCX_{0,2} & (45) \\
 CCX_{0,1} \cdot CCX_{0,2} &\approx CCX_{0,2} \cdot CCX_{0,1} \cdot CCX_{0,2} \cdot CCX_{0,1} & (46)
 \end{aligned}$$

$$X_{[a,b]}^2 \approx \epsilon \quad (1a)$$

$$(-1)_{[a]}^2 \approx \epsilon \quad (1b)$$

$$K_{[a,b,c,d]}^2 \approx \epsilon \quad (1c)$$

$$X_{[a,b]} X_{[c,d]} \approx X_{[c,d]} X_{[a,b]} \quad (2a)$$

$$X_{[a,b]} (-1)_{[c]} \approx (-1)_{[c]} X_{[a,b]} \quad (2b)$$

$$X_{[a,b]} K_{[c,d,e,f]} \approx K_{[c,d,e,f]} X_{[a,b]} \quad (2c)$$

$$(-1)_{[a]} (-1)_{[b]} \approx (-1)_{[b]} (-1)_{[a]} \quad (2d)$$

$$(-1)_{[a]} K_{[b,c,d,e]} \approx K_{[b,c,d,e]} (-1)_{[a]} \quad (2e)$$

$$K_{[a,b,c,d]} K_{[e,f,g,h]} \approx K_{[e,f,g,h]} K_{[a,b,c,d]} \quad (2f)$$

$$X_{[a,d]} X_{[b,b]} \approx X_{[a,d]} X_{[b,d]} \quad (3a)$$

$$X_{[b,b]} X_{[a,b]} \approx X_{[b,b]} X_{[b,b]} \quad (3b)$$

$$X_{[a,b]} (-1)_{[b]} \approx (-1)_{[b]} X_{[a,b]} \quad (3c)$$

$$X_{[a,d]} K_{[a,b,c,d]} \approx K_{[a,b,c,d]} X_{[a,d]} \quad (3d)$$

$$X_{[b,b]} K_{[a,b,c,d]} \approx K_{[a,b,c,d]} X_{[b,b]} \quad (3e)$$

$$X_{[c,c]} K_{[a,b,c,d]} \approx K_{[a,b,c,d]} X_{[c,c]} \quad (3f)$$

$$X_{[d,d]} K_{[a,b,c,d]} \approx K_{[a,b,c,d]} X_{[d,d]} \quad (3g)$$

$$X_{[a,b]} K_{[a,b,c,d]} \approx K_{[a,b,c,d]} X_{[b,d]} (-1)_{[b]} (-1)_{[d]} \quad (4a)$$

$$X_{[b,c]} K_{[a,b,c,d]} \approx (-1)_{[a]} K_{[a,b,c,d]} (-1)_{[a]} K_{[a,b,c,d]} (-1)_{[d]} \quad (4b)$$

$$X_{[c,d]} K_{[a,b,c,d]} \approx K_{[a,b,c,d]} X_{[b,d]} \quad (4c)$$

$$K_{[a,b,c,d]} K_{[b,d,e,f]} \approx K_{[c,d,e,f]} K_{[a,b,c,d]} \quad (5a)$$

$$\begin{aligned}
 (-1)_{[a]} (-1)_{[c]} X_{[a,e]} K_{[c,f,g,h]} K_{[a,b,c,d]} X_{[d,e]} K_{[a,b,c,d]} K_{[c,f,g,h]} X_{[a,e]} (-1)_{[a]} (-1)_{[c]} \\
 \approx
 \end{aligned} \quad (6a)$$

$$K_{[c,f,g,h]} K_{[a,b,c,d]} X_{[d,e]} K_{[a,b,c,d]} K_{[c,f,g,h]} \quad (6a)$$

From W(E8) to O(8, Z[1/2])

From the following section, we have 48 circuit relations for W(E8).

$$\begin{aligned}
 CX_{1,2} &\approx \epsilon & (24) \\
 K_{0,1} &\approx \epsilon & (25) \\
 X_0 &\approx X_0 \cdot CX_{1,2} & (26) \\
 K_{1,2} &\approx K_{1,2} \cdot X_0 & (27) \\
 X_1 &\approx \sigma_{0,1} \cdot X_0 \cdot \sigma_{0,1} & (28) \\
 CX_{2,0} &\approx \sigma_{0,2} \cdot CX_{0,2} \cdot \sigma_{0,2} & (29) \\
 CX_{1,2} &\approx \sigma_{0,1} \cdot CX_{0,2} \cdot \sigma_{0,1} & (30) \\
 CX_{2,1} &\approx \sigma_{0,1} \cdot CX_{2,0} \cdot \sigma_{0,1} & (31) \\
 CCX_{0,1} &\approx \sigma_{0,1} \cdot CCX_{1,2} \cdot \sigma_{0,1} & (32) \\
 CCX_{0,1} &\approx \sigma_{1,2} \cdot CCX_{0,2} \cdot \sigma_{0,1} & (33) \\
 Z_1 &\approx \sigma_{0,1} \cdot Z_0 \cdot \sigma_{0,1} & (34) \\
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 CCX_{0,1} \cdot CCX_{0,2} &\approx CCX_{0,2} \cdot CCX_{0,1} \cdot CCX_{0,2} \cdot CCX_{0,1} & (46)
 \end{aligned}$$

From Li, Ross, and Selinger 2021, we have relation families for all instances of $O(n, Z[1/2])$.

$$\begin{aligned}
 X_{[a,b]} K_{[c,d,e,f]} &\approx K_{[c,d,e,f]} X_{[a,b]} & (2c) \\
 (-1)_{[a]} (-1)_{[b]} &\approx (-1)_{[b]} (-1)_{[a]} & (2d) \\
 (-1)_{[a]} K_{[b,c,d,e]} &\approx K_{[b,c,d,e]} (-1)_{[a]} & (2e) \\
 K_{[a,b,c,d]} K_{[e,f,g,h]} &\approx K_{[e,f,g,h]} K_{[a,b,c,d]} & (2f) \\
 X_{[a,d']} X_{[b,b]} &\approx X_{[a',b]} X_{[a,d]} & (3a) \\
 X_{[b,b']} X_{[a,b]} &\approx X_{[a,b']} X_{[b,b]} & (3b) \\
 X_{[a,b]} (-1)_{[a]} &\approx (-1)_{[a]} X_{[a,b]} & (3c) \\
 X_{[a,d']} K_{[a,b,c,d]} &\approx K_{[a',b,c,d]} X_{[a,d]} & (3d) \\
 X_{[b,b']} K_{[a,b,c,d]} &\approx K_{[a,b,b',c,d]} X_{[b,b]} & (3e) \\
 X_{[c,c']} K_{[a,b,c,d]} &\approx K_{[a,b,c',d]} X_{[c,c]} & (3f) \\
 X_{[d,d']} K_{[a,b,c,d]} &\approx K_{[a,b,c,d']} X_{[d,d]} & (3g) \\
 X_{[a,b]} K_{[a,b,c,d]} &\approx K_{[a,b,c,d]} X_{[b,d]} (-1)_{[a]} (-1)_{[d]} & (4a) \\
 X_{[b,c]} K_{[a,b,c,d]} &\approx (-1)_{[a]} K_{[a,b,c,d]} (-1)_{[d]} K_{[a,b,c,d]} (-1)_{[c]} & (4b) \\
 X_{[c,d]} K_{[a,b,c,d]} &\approx K_{[a,b,c,d]} X_{[b,d]} & (4c) \\
 K_{[a,b,c,d]} K_{[b,d,e,f]} &\approx K_{[c,d,e,f]} K_{[a,b,c,d]} & (5a) \\
 (-1)_{[a]} (-1)_{[c]} X_{[a,c]} K_{[e,f,g,h]} K_{[a,b,c,d]} X_{[d,e]} K_{[a,b,c,d]} K_{[e,f,g,h]} X_{[a,c]} (-1)_{[a]} (-1)_{[c]} & \approx & (6a) \\
 K_{[e,f,g,h]} K_{[a,b,c,d]} X_{[d,e]} K_{[a,b,c,d]} K_{[e,f,g,h]} & &
 \end{aligned}$$

From W(E8) to O(8, Z[1/2])

From the following section, we have 48 circuit relations for W(E8).

$$\begin{aligned}
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 K_{1,2} &\approx K_{1,2} \cdot X_0 & (27) \\
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 CX_{2,0} &\approx \sigma_{0,2} \cdot CX_{0,2} \cdot \sigma_{0,2} & (29) \\
 CX_{1,2} &\approx \sigma_{0,1} \cdot CX_{0,2} \cdot \sigma_{0,1} & (30) \\
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 CX_{0,1} \cdot CZ_{1,2} &\approx CZ_{1,2} \cdot CZ_{0,2} \cdot CX_{0,1} & (43) \\
 CX_{1,2} \cdot CCX_{1,2} &\approx CCX_{1,2} \cdot CX_{1,0} \cdot CX_{1,2} & (44) \\
 CCX_{1,2} \cdot CX_{0,1} &\approx CX_{0,1} \cdot CCX_{0,2} \cdot CCX_{1,2} \cdot CCX_{0,2} & (45) \\
 CCX_{0,1} \cdot CCX_{0,2} &\approx CCX_{0,2} \cdot CCX_{0,1} \cdot CCX_{0,2} \cdot CCX_{0,1} & (46)
 \end{aligned}$$

When $n = 8$, one can calculate that this presentation has 2113 distinct relations.

$$(-1)^{|a|} K_{[a,b,c,d]} \approx K_{[b,c,d,a]} (-1)^{|a|} \quad (2e)$$

$$K_{[a,b,c,d]} K_{[c,f,g,h]} \approx K_{[c,f,g,h]} K_{[a,b,c,d]} \quad (2f)$$

$$X_{[a,d]} X_{[b,b]} \approx X_{[a,b]} X_{[d,d]} \quad (3a)$$

$$X_{[b,b']} X_{[a,b]} \approx X_{[a,b']} X_{[b,b']} \quad (3b)$$

$$X_{[a,b]} (-1)^{|a|} \approx (-1)^{|a|} X_{[a,b]} \quad (3c)$$

$$X_{[a,d]} K_{[a,b,c,d]} \approx K_{[a',b,c,d]} X_{[a,d]} \quad (3d)$$

$$X_{[b,b']} K_{[a,b,c,d]} \approx K_{[a,b',c,d]} X_{[b,b']} \quad (3e)$$

$$X_{[c,c']} K_{[a,b,c,d]} \approx K_{[a,b,c',d]} X_{[c,c']} \quad (3f)$$

$$X_{[d,d']} K_{[a,b,c,d]} \approx K_{[a,b,c,d']} X_{[d,d']} \quad (3g)$$

$$X_{[a,b]} K_{[a,b,c,d]} \approx K_{[a,b,c,d]} X_{[b,d]} (-1)^{|b|} (-1)^{|a|} \quad (4a)$$

$$X_{[b,c]} K_{[a,b,c,d]} \approx (-1)^{|a|} K_{[a,b,c,d]} (-1)^{|a|} K_{[a,b,c,d]} (-1)^{|a|} \quad (4b)$$

$$X_{[c,d]} K_{[a,b,c,d]} \approx K_{[a,b,c,d]} X_{[b,d]} \quad (4c)$$

$$K_{[a,b,c,d]} K_{[b,d,c,f]} \approx K_{[c,d,e,f]} K_{[a,b,c,d]} \quad (5a)$$

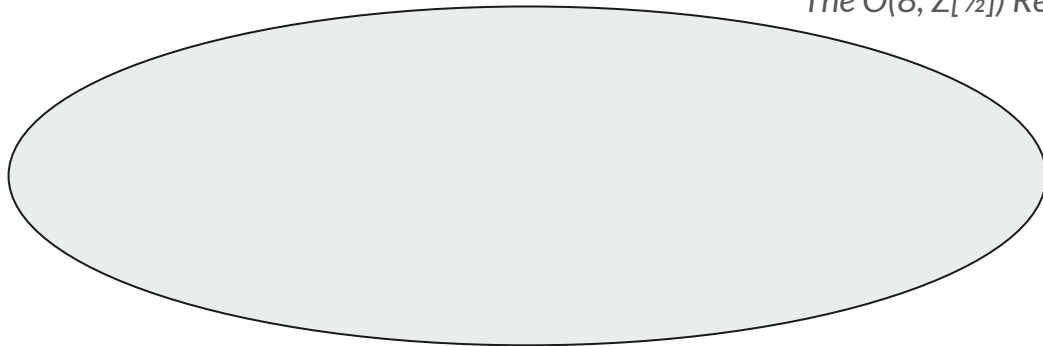
$$\begin{aligned}
 (-1)^{|a|} (-1)^{|c|} X_{[a,c]} K_{[c,f,g,h]} K_{[a,b,c,d]} X_{[d,e]} K_{[a,b,c,d]} K_{[c,f,g,h]} X_{[a,c]} (-1)^{|a|} (-1)^{|c|} \\
 \approx
 \end{aligned} \quad (6a)$$

$$K_{[c,f,g,h]} K_{[a,b,c,d]} X_{[d,e]} K_{[a,b,c,d]} K_{[c,f,g,h]}$$



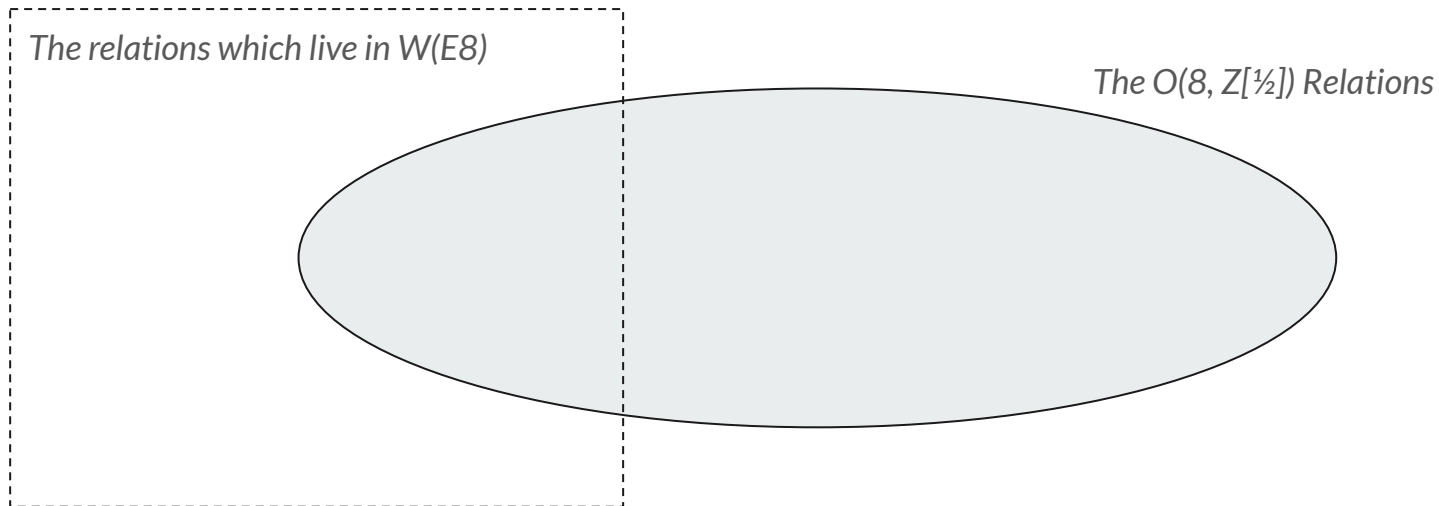
Proof Idea: Extending the Presentation

The $O(8, \mathbb{Z}[\frac{1}{2}])$ Relations

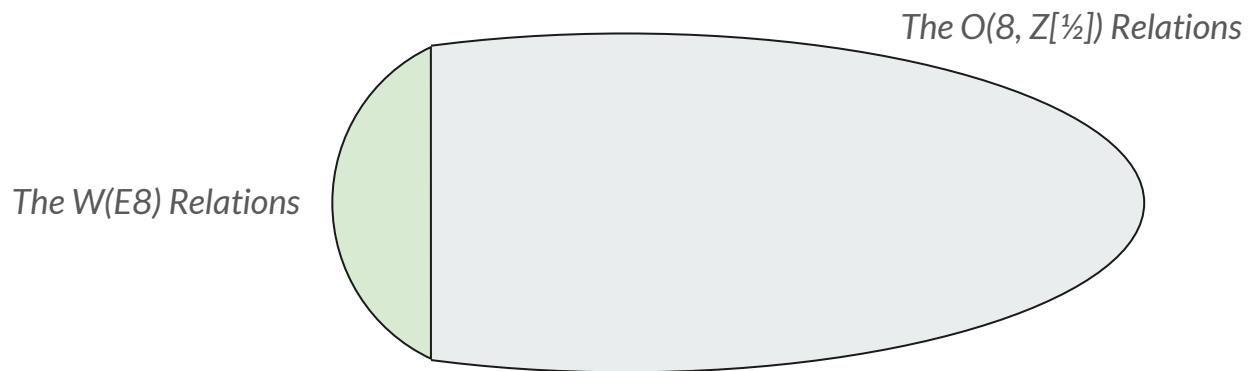




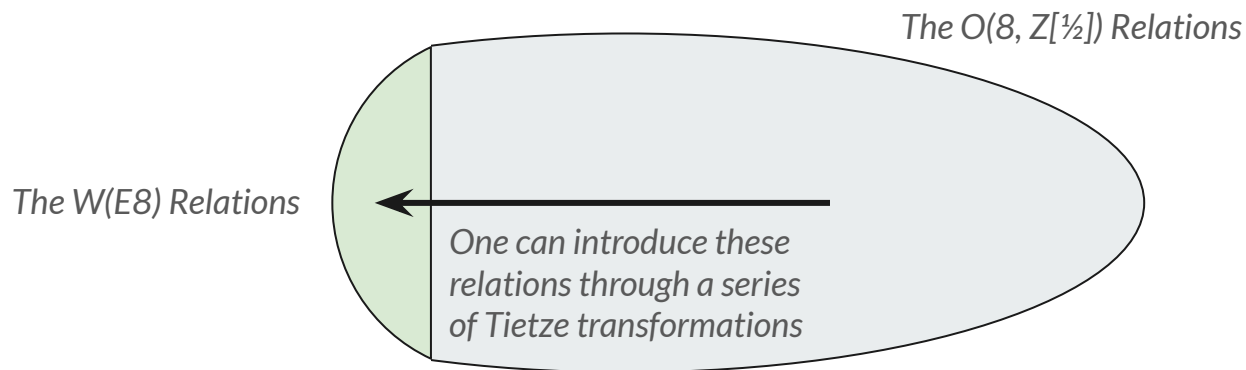
Proof Idea: Extending the Presentation



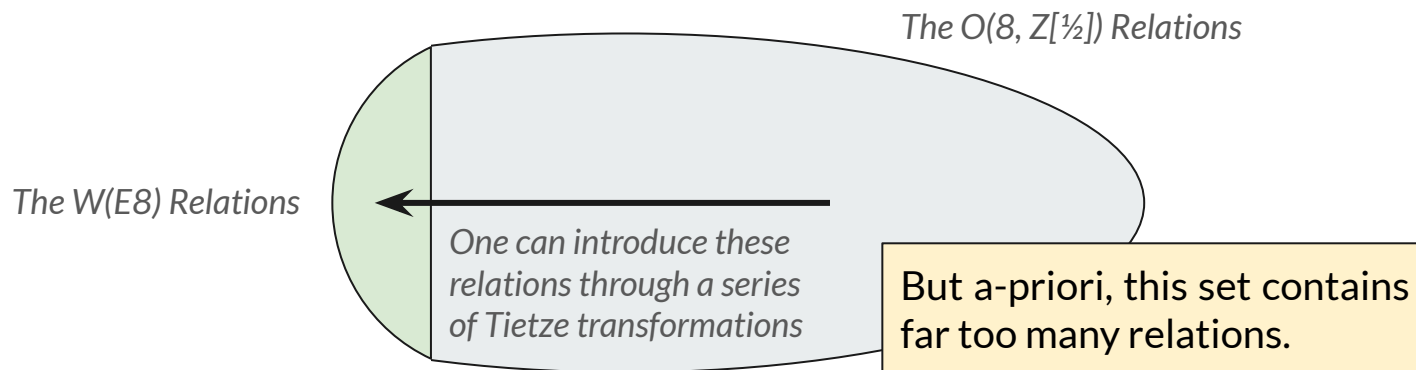
Proof Idea: Extending the Presentation



Proof Idea: Extending the Presentation



Proof Idea: Extending the Presentation





Reducing the Number of $O(n, Z^{[1/2]})$ Relations

Reducing the Number of $O(n, \mathbb{Z}[\frac{1}{2}])$ Relations

1. Deriving the Braid Relations (from $W(E_8)$)

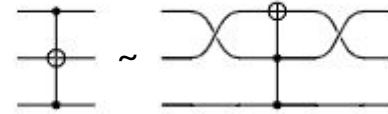


Reducing the Number of $O(n, \mathbb{Z}[\frac{1}{2}])$ Relations

1. Deriving the Braid Relations (from $W(E_8)$)



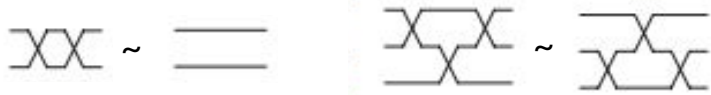
2. Deriving Gate Permutations (using $W(E_8)$)



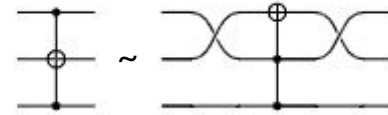
$\sim$

Reducing the Number of $O(n, \mathbb{Z}[1/2])$ Relations

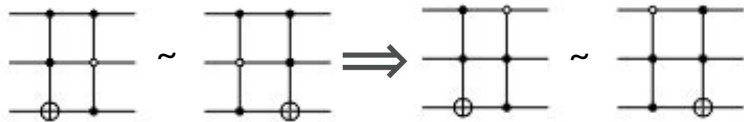
1. Deriving the Braid Relations (from $W(E_8)$)



2. Deriving Gate Permutations (using $W(E_8)$)



3. Deriving Instances of Relation Families

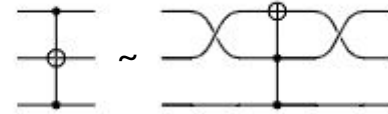


Reducing the Number of $O(n, \mathbb{Z}[\frac{1}{2}])$ Relations

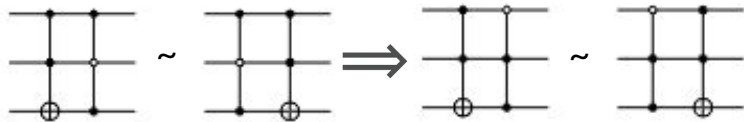
1. Deriving the Braid Relations (from $W(E_8)$)



2. Deriving Gate Permutations (using $W(E_8)$)



3. Deriving Instances of Relation Families

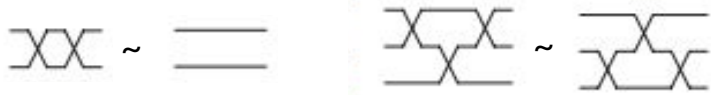


4. Deriving Redundant Relations

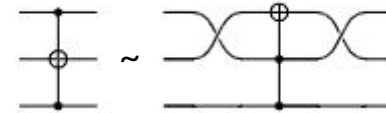
*Several families of relations are redundant, and may be eliminated entirely via a sequence of **Rel**(-) transformations.*

Reducing the Number of $O(n, \mathbb{Z}[\frac{1}{2}])$ Relations

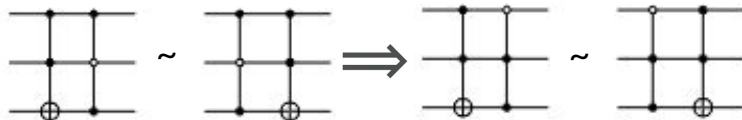
1. Deriving the Braid Relations (from $W(E_8)$)



2. Deriving Gate Permutations (using $W(E_8)$)



3. Deriving Instances of Relation F



This reduces the case where $n = 8$ from 2112 relations to only **53 relations**.

Redundant Relations

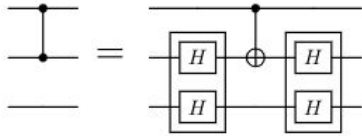
Several families of relations are redundant, and may be eliminated entirely via a sequence of **Rel(-)** transformations.

The Full Presentation

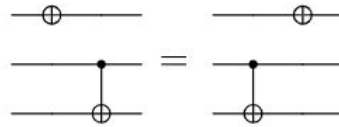
Qualitative and Quantitative Summary

Classification: A Qualitative Summary

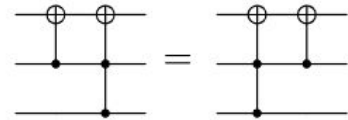
Alias Relations



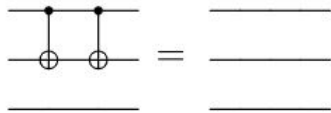
Bifactoriality Relations



Commutativity Relations



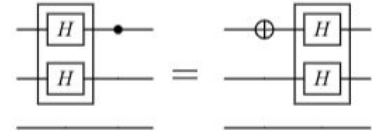
Order Relations



Symmetry Relations

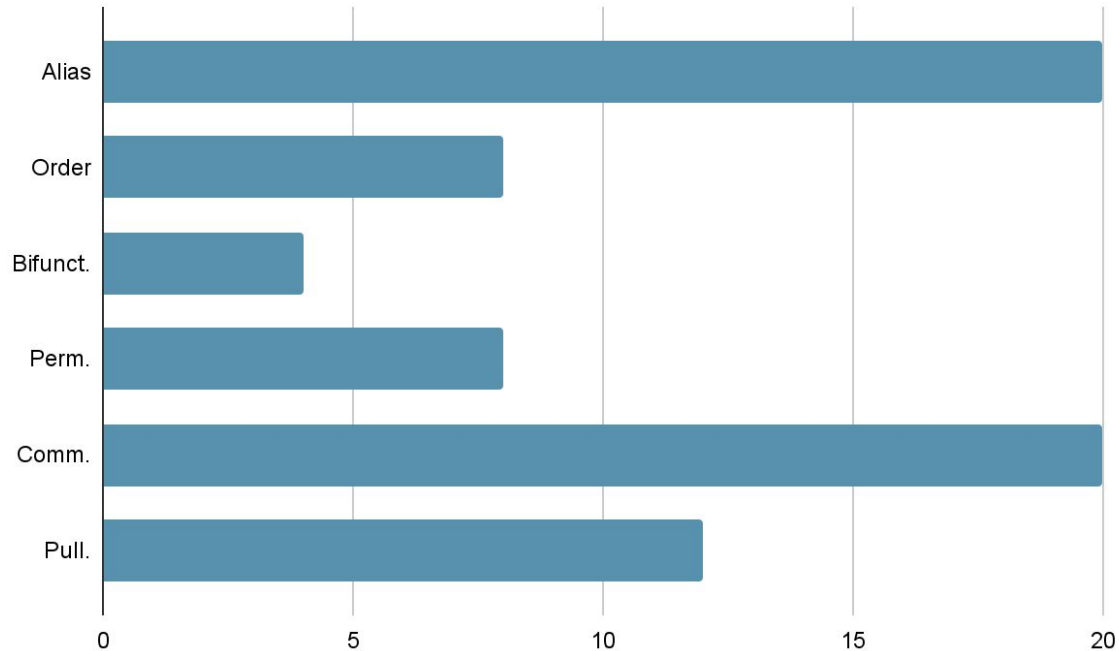


"Pulling Through"





A Quantitative Summary of Each Relation Class



Across all six classes, there are 72 distinct relations for the general case.



Directions for Future Work

1. What is a presentation for 4-qubit (or n -qubit) Toffoli-Hadamard circuits.
2. Do the dyadic 3-qubit Toffoli-Hadamard circuits form an abstract Coxeter group?
3. What can be said about the Toffoli-complexity for 3-qubit Toffoli-Hadamard circuits?

Thank You For Listening!
Questions? Comments?

